

Reskilling and Upskilling in the AI Era

Lai Jingting

p127629@siswa.ukm.edu.my

*Faculty of Economic and Management,
National University of Malaysia, School of Architectural Engineering, Malaysia
Nanning Vocational and Technical University, China
<https://orcid.org/0009-0001-9517-8113>*

Elaina Rose Johar

elainarose@ukm.edu.my

*Faculty of Economic and Management,
National University of Malaysia, Malaysia
<https://orcid.org/0000-0002-9756-0228>*

Yin Junjia

18908453651@163.com

*Department of Civil Engineering, Faculty of Engineering,
Universiti Putra Malaysia, Malaysia
<https://orcid.org/0000-0003-0581-0603>*

Jiacheng Zheng

jiacheng.james.zheng@outlook.com

*Department of Natural Science and Mathematics,
University of Denver, America*

Corresponding Author: Yin Junjia

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Abstract

In an era of AI-driven disruption, reskilling and upskilling have become key levers for maintaining workforce relevance. Recent bibliometric data show an annual growth rate of 23% in related research since 2022, a trend that is in sync with the rise of generative AI technologies. This study uses bibliometric analysis based on VOSviewer to map the evolution of relevant research topics in this field from 2015 to 2025. Through the literature indexed by Scopus, this paper finds that there is a "core-periphery" structure of knowledge production in this field, with many high-impact studies coming from developed countries, obscuring the contextual approaches of developing economies. Current research focuses on key clusters including corporate digital training, policy-led initiatives and ethical frameworks, and personalized learning with artificial intelligence. The analysis highlights systemic gaps, such as the underrepresentation of gender-inclusive strategies and the lack of longitudinal studies tracking policy effectiveness beyond short-term interventions. We propose three future directions: (a) integrating global South perspectives through databases such as MyCite and the ASEAN Citation Index; (b) adopting mixed methods that combine quantitative bibliometrics with qualitative case studies; and (c) developing standardized metrics to assess the equity impact of reskilling programs. This study not only maps the knowledge structure of the field, but also critiques its geographical and methodological biases, calling for a more inclusive research agenda to address the growing skills gap in the AI economy.

Keywords: Reskilling, Upskilling, AI adoption, Bibliometric analysis, VOSviewer.

1. INTRODUCTION

Reskilling and upskilling are crucial in rapid technological advancement and digital transformation [1]. The terms describe developing new competencies or refining existing ones to align with the evolving demands of the labor market and various industries [2]. The integration of new technologies has transformed operations into 86% of firms. This is expected to reshape workplace competencies comprehensively, influencing skills, knowledge, capabilities, and attitudes as employees adjust to increasing automation and Artificial Intelligence (AI) [3]. Around 40% of European executives surveyed stated that there was a lack of employees with the skills needed to use new technologies [4]. Reskilling and upskilling the workforce is Malaysia's focus, which is to meet challenges driven by the rapid evolution of technology, particularly technological innovations in AI. Malaysia Human Resource Minister V Sivakumar touched upon the estimated 4.5 million Malaysian job losses by 2030 unless the workers adjust [5]. The Malaysian government has stressed the need for continuous skill development among workers to remain competitive [6].

Employee upskilling and reskilling are of cardinal importance to workforce development during digitalization and Industry 4.0. Current literature stresses employee reskilling and upskilling in implementing AI and aligning HR practices with Industry 4.0 principles is particularly important [7]. This topic is particularly pertinent in emerging economies such as India, where the challenge of sustainable human resource management is posed by disruptions caused by Industry 4.0 and must be addressed. The employees' mindset towards adopting AI in industries like the Indian banking sector is instrumental in successful implementation [8]. Even though AI is rapidly developing, regions such as ASEAN do not yet possess a complete governance structure to contend with challenges imposed by the implementation of AI [9]. Research shows that the need exists for employees to reskill and upskill and thereby integrate into the workplace use of AI effectively, in addition, existing studies have examined the impact of AI adoption on dynamic capabilities, employee engagement and organizational agility [10], job engagement [11], employee trust [12], and organizational agility [13]. To contend with challenges posed to work by AI-mediated automation, nations, educators, and corporations need to come together to equip employees with the required training and assistance [14]. There remains significant potential to extend bibliometric studies explicitly focused on employee reskilling and upskilling within AI integration. While earlier studies have utilized bibliometric methods to investigate employee training or skill development themes, few of them capture the nexus of reskilling/upskilling and AI in as much depth and complexity as required. As businesses become more engaged in adopting artificial intelligence in their operations, technological skill shortcomings and managerial and cross-cultural barriers to change have become of interest.

This research integrates a range of reliable bibliometric indicators from author co-citation and institutional productivity to keyword evolution and geographic production. It offers further and brighter light on the scholarship in the area. This research examines in-depth earlier papers on reskilling and upskilling employees developed in the Scopus database to build a bibliometric profile of the field and determine likely research directions in the future. It offers researchers and practitioners a complete picture of existing research around employee reskilling and upskilling, focusing on major studies, renowned authors, prominent affiliations, core concepts, and methods in the research area.

Identified insights and methods can be helpful to organizations and policy-making bodies to enhance employee training, learning, and development.

2. LITERATURE REVIEW

“Upskilling” concerns acquiring new skills and improving existing skills that align with an employee’s current job and a company in the industry [15]. Up-skilling is concerned with improving the skills one must perform in their existing job [16], whereas reskilling is acquiring new capacities to perform in other job functions or roles. Upskilling attempts to boost one’s employment prospects or enhance job performance [17]. The literature highlights the importance of reskilling and upskilling in various sectors. Li [1] outlines a shift in required skills across different job sectors, driven by integrating AI, machine learning, and other disruptive technologies. The need for reskilling is a global issue, affecting both developed and developing economies. A separate study examining upskilling and reskilling in the logistics and supply chain sectors, specifically during the Fourth Industrial Revolution, revealed that nearly 70% of respondents felt the training offered by their organizations had broadened professional opportunities within their fields [18]. Leon [19] contends that upskilling and reskilling enable employees to grow, fostering a sense of progress and value within the organization. Such practices can also help retain talent by reducing redundancies and outsourcing, strengthening the organization’s resilience and agility. Consequently, organizations should continuously assess the need for skill development and actively encourage employees to acquire new competencies [20].

Previous bibliometric analyses on reskilling and upskilling have highlighted specific issues, including but not limited to employee reskilling and upskilling [21]. Similarly, an earlier study by Achoki [22] focused on the same research area, although it was restricted to using VOSviewer and the Scopus database for analysis. However, contrasting perspectives in the literature emphasize the challenges associated with reskilling and upskilling in practice. Murat and Lawton [20] argue that many upskilling initiatives fail due to a lack of contextual relevance and managerial resistance, while Angrave et al. [23] highlight the limitations of standardized digital learning platforms that overlook industry-specific needs. These findings indicate that although theoretical support for skill development is robust, there is a need for a more nuanced understanding of its real-world implementation.

3. MATERIALS AND METHODS

Grounded initially in information science, bibliometric analysis has gained substantial popularity and broadened its scope to include cross-disciplinary applications, especially within business research [24]. Furthermore, bibliometric analysis is not merely a trending approach among researchers. It demonstrates its utility in managing vast scientific data and generating significant research impact [25]. Bibliometric analysis has gained widespread popularity in library and information sciences, where relevant tools in statistics are utilized to assess and evaluate existing academic publications. However, its use in business research is still relatively new, and in some instances, its full potential has yet to be fully realized [26]. These tools empower researchers to pinpoint research clusters, acquire more profound understandings of prevailing research themes,

and uncover emerging directions within a specific area. A four-step approach was employed following a systematic literature search, incorporating bibliometric citation analysis to identify the most influential and frequently cited journals, papers, affiliations, authors, and keywords (See Figure 1).

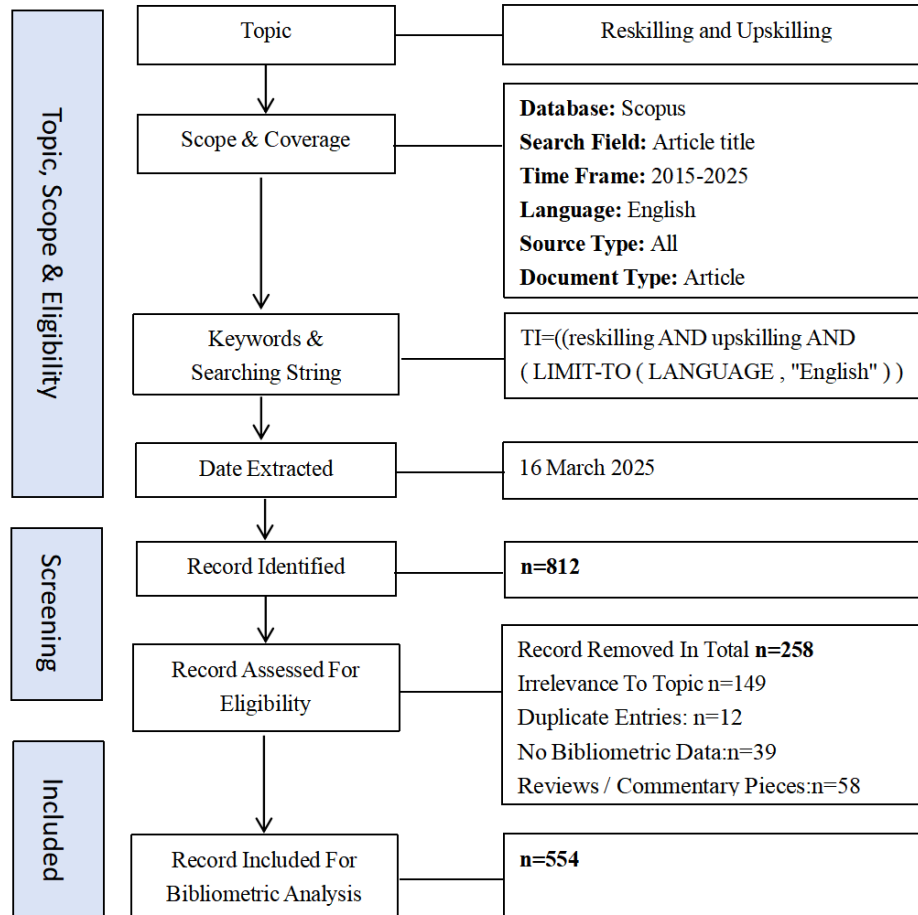


Figure 1: Search process and content.

This study utilized the Scopus database to gather data from journal articles for bibliometric analysis. Scopus is a widely used database for evaluating scientific journals [27]. Scopus has been extensively analyzed through bibliometric methods across various disciplines [28]. It is regarded as a top-tier authoritative database for high-quality academic articles. The reliability of Scopus as a structured source of high-quality bibliometric data has led to its widespread adoption in literary research, particularly in quantitative scientific research and studies [29]. The comparative analysis of journal coverage in Scopus, Web of Science (WoS) and Dimensions reveals significant overlap between these databases, with Scopus indexing a substantial portion of the journals also covered by Web of Science [30]. This study adopted a systematic search strategy within the Scopus database to ensure methodological rigor. The search was conducted using article titles of journal papers published between 2015 and 2025. Only peer-reviewed English-language journal articles were included. Preferred Reporting Items for Systematic Review and Meta-analysis review method, also known as the PRISMA, was used to gather articles from Scopus on reskilling and upskilling, consisting of

three stages: identification, screening, and inclusion/exclusion criteria. The PRISMA framework was applied to ensure transparency and replicability throughout the selection process.

The literature was systematically examined in the first stage using the Scopus database and the advanced search option to identify relevant terms. To narrow the focus of the study, the following search string was applied, with the research themes being the primary topics of interest: TI=(reskilling AND upskilling AND (LIMIT-TO (LANGUAGE, "English"))). The preliminary search resulted in 812 articles, which were then refined through the selection and elimination procedure (Table 1). In the second phase, these articles were analyzed based on various criteria, such as (1) Publication Year, (2) Document types, (3) Scopus categories, and (4) Language. A total of 258 records were removed during the screening and eligibility assessment process. Specifically, 149 records were excluded due to irrelevance to reskilling and upskilling. Additionally, 12 duplicate entries were identified and removed. 39 records lacked bibliometric data, making them unsuitable for inclusion in the analysis. Lastly, 58 records were excluded as they were either review articles or commentary pieces rather than original research containing bibliometric information. Articles were excluded due to a lack of relevance to core themes or as duplicate entries, despite keyword matches in titles. After applying these exclusion criteria, 554 records were deemed eligible and included for bibliometric analysis using VOSviewer. The study elaborately selected English research articles indexed in the Scopus database for the bibliometric analysis.

Table 1: Inclusion and Exclusion of Data

Records	Included
Publication Year	2015-2025
Document types	Article, book chapters, editorial material
Scopus categories	Social Sciences; Business, Management and Accounting; Computer Science; Engineering; Economics, Econometrics and Finance; Decision Sciences; Psychology; Arts and Humanities
Language	English

The study contains articles from various fields such as social sciences, business management, accounting, computer science, engineering, economics, econometrics, finance, decision sciences, arts and humanities, and interdisciplinary studies to ensure the relevance of data related to reskilling and upskilling. It focused on papers published within the past decade (2015 to 2025) to maintain the subject's quality, conciseness, and relevance. This approach resulted in the inclusion of 554 articles for bibliometric analysis. VOSviewer, a tool designed to create and interpret bibliometric maps efficiently, is increasingly utilized in bibliometric research due to its intuitive interface and robust analytical capabilities [31, 32]. The approach supports the organized gathering of applicable literature. It helps identify and visualize relationships among selected publications and provides deeper insights into the organization and dynamics of research areas [33]. The interdisciplinary nature of this study—spanning social sciences, business, engineering, and humanities—reflects the multifaceted reality of reskilling and upskilling in the digital age. However, this breadth also raises methodological challenges. While bibliometric tools like VOSviewer excel at mapping co-citation networks and keyword trends, they may overlook nuanced disciplinary differences in how “skills” are conceptualized. For instance, a computer science paper on AI upskilling likely employs different frameworks than an economics study on labor market reskilling policies. The risk is that clustering algorithms prioritize quantitative connections (e.g., citation counts) over qualitative

alignment, potentially obscuring fragmented or contested theories. Furthermore, the 2015-2025 timeframe captures Industry 4.0's rise but may miss early signals of *Industry 5.0* trends (e.g., human-AI collaboration skills). Future research could employ predictive bibliometrics to identify emerging themes—such as “green skills” for sustainability or “metaverse literacy”—before they reach critical mass.

4. ANALYSIS AND RESULTS

4.1 Descriptive Analysis

The research carries out a complete bibliometric survey of research citing reskilling and upskilling published in the past decade from 2015 to 2025. The survey aims to reveal meaningful patterns and trends in the growing field of scholarly research in reskilling and upskilling, especially in response to rapid technological innovation and shifting labor market demands in the context of the Fourth Industrial Revolution and artificial intelligence. 554 peer-reviewed papers were determined to be relevant and were used in the analyzed dataset at the end. These papers were spread over 412 unique scholarly journals and other publications, indicative of a broad interdisciplinary interest in the topic among various disciplines ranging from education and human resource management to economics, technology studies, and labor market policy. This spread of source publications indicates that the scope of the literature on reskilling and upskilling is multidisciplinary. Out of 554 selected papers, 1,795 individual scholars were involved in writing these papers, which reflects a high tendency for collaboration in the field. On average, each document received approximately 5.41 citations, suggesting moderate scholarly attention and impact. Regarding authorship patterns, most of the contributions came from multi-authored works, highlighting the collaborative research culture prevalent in this domain. Notably, 96 authors contributed to single-authored publications, reflecting a small but significant portion of independent scholarly engagement with the topic.

4.2 Main Information

Table 2 shows that the academic focus on reskilling and upskilling first emerged in 2015 with a foundational paper in the Scopus database, drawing from a wide range of sources. Since 2020, however, publication activity has surged dramatically, with a 116.9% increase over five years—a trend driven by accelerating technological disruption, pandemic-induced labor shifts, and the growing urgency of workforce adaptation. The field exhibits characteristics of an evolving discipline: articles average 5.41 citations (2.695 annually), while the low average publication age (1.65 years) suggests newer studies are still gaining traction. Collaboration dominates with only 17% single-authored works and a high co-authorship ratio (3.7 per document), indicating interdisciplinary engagement. Yet, fragmented contributions (0.30 documents per author across 1,795 researchers) reveal a dispersed research landscape. From the perspective of the profound logic of bibliometrics, this research field presents a typical “paradox of emerging discipline development.” On the one hand, the co-authorship rate is as high as 3.7 and cross-institutional cooperation is common, indicating a high degree of consensus in the academic community on the topic of skill remodeling. On the other hand, the scattered state of each researcher contributing only 0.3 articles on average exposes the lack of research depth and continuity. This contradiction reflects the multi-dimensional nature of

skills research in the context of digital transformation—the speed of technological iteration forces research to maintain breadth but lacks the theoretical accumulation of classical disciplines. There is an apparent disconnect between the 100% single publication form of journal papers and the actual demand of the labor market: when 70% of Southeast Asian enterprises urgently need digital transformation talents, the academic community rarely provides practical skills assessment frameworks or training program verification. Future research may need to build a “triple helix” knowledge production model, by establishing a collaborative laboratory between enterprises, governments and universities.

Table 2: Main Information of the Selected Articles

Types	Description	Results
Main information of data	Timespan	2015:2025
	Sources	412
	Documents	554
	Average age of publications	1.65
	Average citations per document	5.41
	Average citations per year per document	2.695
	References	33756
Document contents	Keywords plus	2515
	Author's keywords	1685
Authors	Number of authors	1795
	Author appearances times	1873
	Authors of single-authored documents	96
	Authors of multi-authored documents	458
Authors collaboration	Single-authored documents	96
	Documents per author	0.30
	Authors per document	3.24
	Co-authors per document	3.7
	Collaboration index	0.83

4.3 Annual Publication Patterns

Figure 2 displays the annual publication volume from the years 2015 to 2025. The bibliometric data reveal that in 2022 and 2023, approximately 104 and 126 articles were published on reskilling and upskilling, respectively. 157 more articles, by 2024, were published in Reskilling and upskilling. Therefore, it is highly anticipated that the quantity of articles on reskilling and upskilling will continue to rise in 2025. Notably, within the first three months of 2025, 63 publications have already been recorded, representing a dramatic surge in scholarly output. This trend suggests a strong likelihood that the total number of publications will surpass previous years by 2025. Given the increasing attention to reskilling and upskilling across various industries. The pronounced growth in publication output not only signals the growing academic recognition of the topic's relevance but also suggests that reskilling and upskilling will remain as a central theme in further study, particularly within the broader discourse on digital labor, human capital development, and sustain-

able employment in the AI-driven economy. The exponential growth from 2023 to 2025 (with an average annual growth rate of 23%) is highly consistent with the popularity curve of generative AI such as ChatGPT, suggesting that technological mutations have moved from theoretical discussion to large-scale practical response. The motivation behind this data is threefold: first, the “creative destruction” effect of AI on the labor market is forcing academics to accelerate their search for solutions. Second, the policy competition between developing countries (e.g., Malaysia’s MyMahir program) and developed countries (the EU’s “Digital Europe Initiative”) has given rise to a large number of localized case studies; moreover, the cooperation between education technology companies (Coursera, Udacity) and academic institutions has made it possible to obtain a window for empirical research on new training modes, such as micro-credentials and personalized learning with AI. However, two pitfalls need to be guarded against in the academic boom.

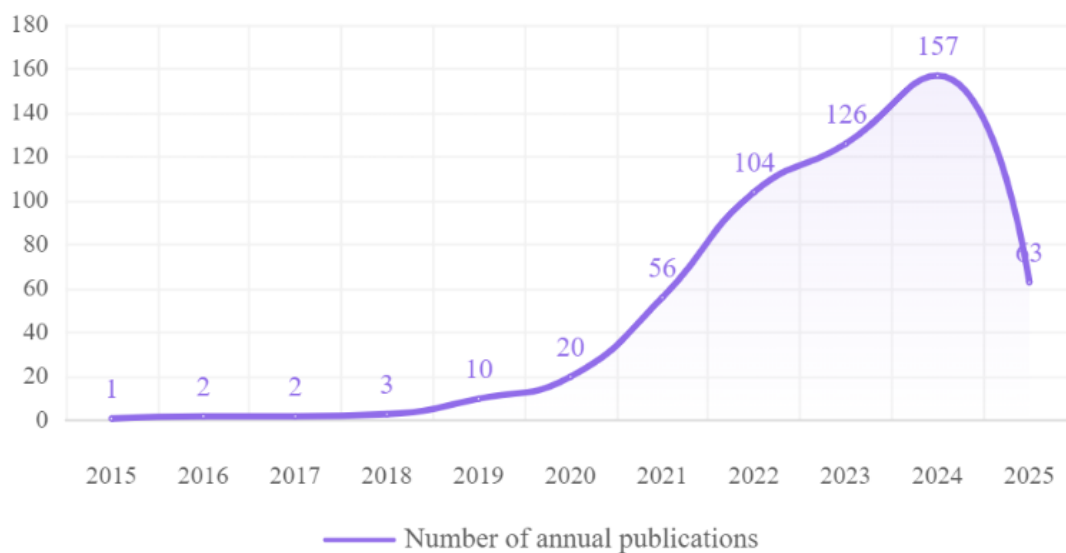


Figure 2: Annual number of publications from 2015 to 2025.

4.4 Most Productive Authors

The authorship data was extracted from Scopus and processed in Microsoft Excel for detailed analysis. The examination in Excel focused on categorizing articles based on authorship type, explicitly identifying the number of publications with single author versus ones with joint authors [34]. Figure 3 presents the top 20 most active authors from 2015 to 2025. It can be concluded that Andrew Ebekozen and Jordi Diaz are the top active authors in reskilling and upskilling, both having published six articles. Other highly cited authors include Daphne Halkias, Hazel Gruenewald, and Jonathan Westover, who have published four articles. These authors have likely contributed foundational studies or frequently referenced frameworks relevant to workforce development, Industry 4.0, and digital transformation. The prominence of multiple co-authored publications among the top contributors suggests that collaboration and interdisciplinary research are significant drivers of

impact in this field. Based on the articles with the most citations, the research of Vallor [35] and Li [1] primarily concentrated on reskilling and upskilling, particularly concerning the roles of ICTs and Industrial 4.0. Mukherjee et al. [36], Raja and Muthuswamy [37] also highlighted Industrial 4.0 and 5.0 skills. Furthermore, Adami et al. [38] suggested that educating and reskilling/upskilling engineering workers in highly technological environments is necessary. Zhu et al. [39] indicated that reskilling and upskilling are widely applied in areas regarding the application of Robotics, AI, and Automation (RAIA). Several articles also highlighted that AI in the workplace significantly requires worker-wise reskilling and upskilling [40]. From the perspective of the deep mechanism of academic production and knowledge dissemination, the current distribution of author influence shows an apparent “Matthew effect” - a few core authors have formed the “gatekeeper” role of academic discourse through continuous output.

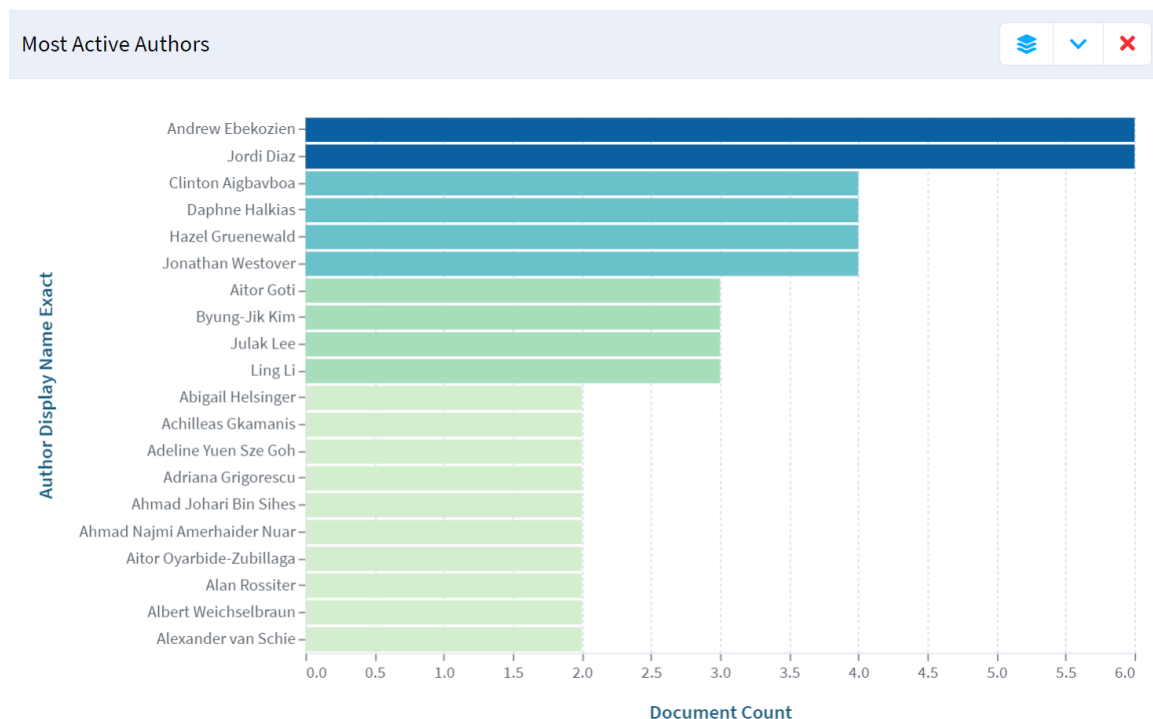


Figure 3: Top 20 most active authors from 2015 to 2025.

Table 3 reveals the triple paradigm shift in skill reshaping research: First, the highly cited literature (such as Jaiswal et al. [41]) all points to the dialectical relationship between technological paradigm change and labor capacity reconstruction. Among them, the “Future-Ready Workforce Development for Industry 4.0” with 152 citations established the theoretical framework of “technology-skill co-evolution”. In comparison, the research on AI skills of multinational companies (147 citations) verified the “dynamic matching hypothesis” between skill supply and enterprise digital transformation. Second, Wach et al. [42]’s critical research on generative AI risks (137 citations) and Vallor’s [43] discussion on technology ethics (122 citations) constitute the emerging “technological disruption-humanistic reflection” research axis, reflecting the academic community’s trend of revising technological optimism. Third, the parallel selection of Industry 5.0 literature (Raja

Table 3: Most Cited Articles on Reskilling and Upskilling

No.	Document Title	Authors	Journal Title	Citations
1	Reskilling and Upskilling the Future-ready Workforce for Industry 4.0 and beyond.	Li [1]	Information System Front	152
2	Rebooting employees: Upskilling for artificial intelligence in multinational corporations	Jaiswal et al. [41]	Artificial Intelligence and International HRM	147
3	The dark side of generative artificial intelligence: A critical analysis of controversies and risks of ChatGPT	Wach et al. [42]	Entrepreneurial Business and Economics Review	137
4	Moral deskilling and upskilling in a new machine age: Reflections on the ambiguous future of character	Vallor [43]	Philosophy & Technology,	122
5	Industry 5.0 or industry 4.0S? Introduction to industry 4.0 and a peek into the prospective industry 5.0 technologies	Raja and Muthuswamy [37]	International Journal on Interactive Design and Manufacturing	115
6	Effectiveness of VR-based training on improving construction workers' knowledge, skills, and safety behavior in robotic teleportation	Adami et al. [38]	Advanced Engineering Informatics	97
7	A comprehensive systematic review of MOOC research: Research techniques, topics, and trends from 2009 to 2019	Zhu et al. [39]	Educational Technology Research and Development	77
8	Employees' perceptions of the implementation of robotics, artificial intelligence, and automation (RAIA) on job satisfaction, job security, and employability	Bhargava et al. [44]	Journal of Technology in Behavioral Science	70
9	Worker and workplace Artificial Intelligence (AI) coexistence: Emerging themes and research agenda	Zirar et al. [40]	Technovation	61
10	The impact of artificial intelligence on workers' skills: Upskilling and reskilling in organizations	Morandini et al. [12]	Informing Science	57

and Muthuswamy [37]) and VR training research (Adami et al. [38]) indicates that the research focus has deepened from the technology application level to the cognitive dimension of “human-AI symbiosis”. The citation data exposes three methodological limitations that need to be overcome: First, only 29% of the top 10 papers adopt a longitudinal research design (such as Zirar et al. [40]), and most rely on cross-sectional data, which makes it challenging to capture the dynamic characteristics of the skill half-life; second, the organizational skill mapping model proposed by Morandini et al. [12] (57 citations) is groundbreaking, but has not yet been combined with neuroscience-driven personalized learning assessment; third, there is a clear “education-employment” data gap between MOOC research (Zhu et al. [39]) and workplace RAIA research (Bhargava et al. [44]), exposing the existing research’s neglect of the real-time feedback mechanism of “lifelong learning-

job requirements”. From the perspective of the evolution of research paradigms, these three shifts reflect that skill research is undergoing a cognitive leap from “technology adaptation theory” to “human-centered theory”. It is particularly noteworthy that there is a fundamental contradiction in the current research at the methodological level: on the one hand, it emphasizes the dynamic nature of skill formation (such as the half-life of skills), but on the other hand, it generally adopts a static research design. This “time and space mismatch” makes it difficult for research results to guide the ever-changing workplace practice.

Based on citation network analysis, we propose three frontier directions: 1) “skill quantization” measurement framework - drawing on Adami et al.’s [38] VR skill acquisition research, developing a real-time ability assessment system that integrates bio signal recognition (such as EEG) and behavioral analysis; 2) “negative skills” research - extending Vallor’s [43] ethical discussion, it is necessary to quantify the degradation of moral ability caused by automation (such as AI dependence weakening crisis decision-making); 3) “skill blockchain” ecosystem construction - integrating Jaiswal et al.’s [41] multinational enterprise research, exploring a global skill certification system based on smart contracts, and solving the problem of cross-border skill recognition in the era of Industry 5.0 (Raja and Muthuswamy [37]). These directions point to a core proposition: when the skill update cycle is shorter than the traditional education iteration cycle, how can we reconstruct the underlying logic of human capital development?

4.5 Visualization and Mapping

The collected bibliometric data was visualized by VOSviewer, a software tool for creating and interpreting bibliometric network maps. The visualizations were presented as density maps or network visualizations with graphical elements, such as color, circle size, font size, and the thickness of connecting lines [45]. For instance, alike colors highlighted clusters or groups, which could correspond to countries or authors with thematic or collaborative relationships. The thickness of strands represented the bond of correlation or collaboration between units (e.g., countries), quantified as relative strength; greater thickness denoted stronger cooperation. Similarly, larger circles or font sizes signify higher levels of productivity or impact.

The visualization of keywords related to reskilling and upskilling from 554 academic papers, as generated using VOSviewer, reveals significant thematic clusters and their interconnections within the research domain. VOSviewer can generate three different types of mapping visualizations: the network graph shown in Figure 4, the overlay graph in Figure 5, and the density graph in Figure 6. This visualization offers a thorough overview of academic literature, highlighting critical areas. The strong connections between AI, education, and sustainability suggest an interdisciplinary approach to workforce transformation. In the central Themes, Artificial Intelligence (AI) emerges as the central and most prominent keyword, represented by the largest node. This signifies its foundational role in reskilling and upskilling research, particularly in driving technological transformations. Figure 4 shows the VOSviewer-generated overlay visualization that identifies the evolution and intersections of prominent research topics in reskilling and upskilling in 554 papers.

From the perspective of knowledge geography, the distribution pattern of countries/regions presented in Figure 5 deeply reveals the phenomenon of “knowledge power asymmetry” in global skill reshaping research. The United States leads with an absolute advantage of 79 papers and 711

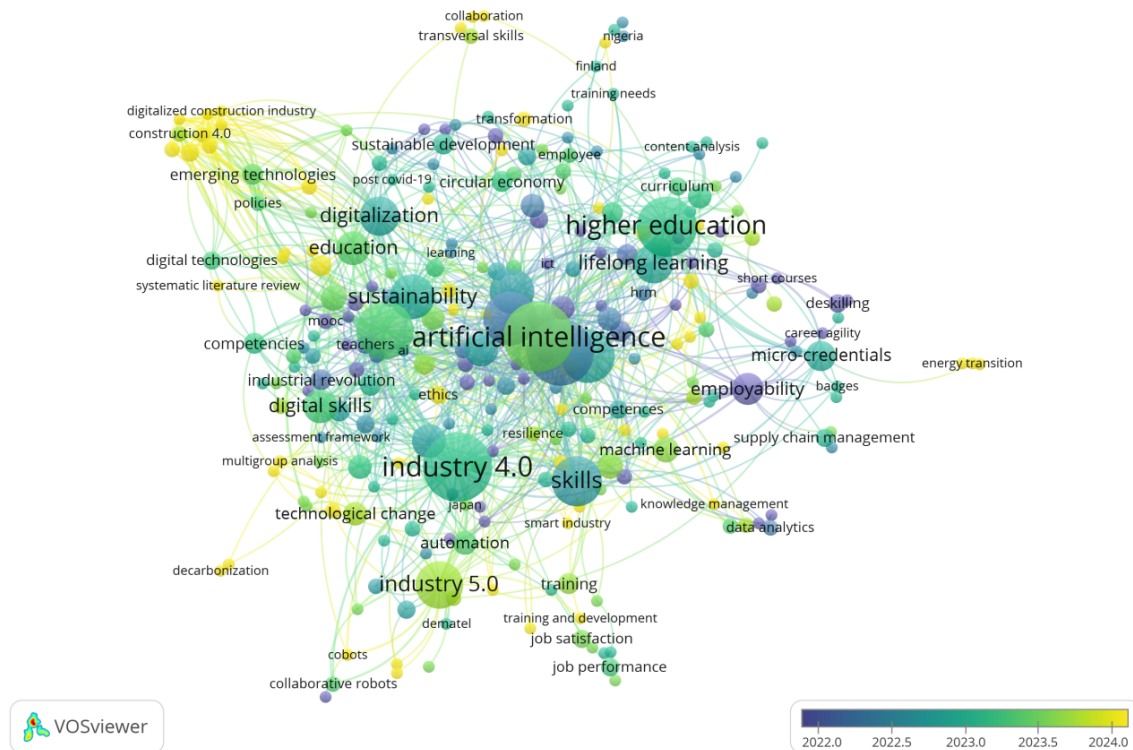


Figure 4: Overlay visualization of keywords.

citations, which not only reflects the research vitality of its academic system but also implies the concentration of global skill discourse power - the “future skills” framework defined by the American academic community (such as STEM priority and Silicon Valley model) is shaping global standards through high-impact journals. It is interesting to note that although Germany and China are both manufacturing powerhouses, their publication volume (18 papers each) is seriously mismatched with their industrial strength, which may reflect two problems: first, the research results of non-English-speaking countries have “visibility loss” in databases such as Scopus; second, application-oriented skill innovation (such as Germany’s dual system upgrade and China’s industry-education integration) has not yet been fully transformed into academic discourse. More seriously, only South Africa has a significant output (25 papers) in Africa, exposing the “missing South” in the global skill research landscape. Future research needs to break through three major bottlenecks: 1) Establish a multilingual literature tracking mechanism to capture the localization practices of non-English speaking countries; 2) Develop “industry-academic” coupling indicators to measure the response speed of research results to fundamental skill gaps; 3) Introduce geo-economic variables (such as automation substitution rates of various countries) in knowledge graph analysis to reveal the deep motivations of research bias differences between countries. This critical reflection may promote the value reconstruction of skill research from “academic influence competition” to “solving actual inequality.”

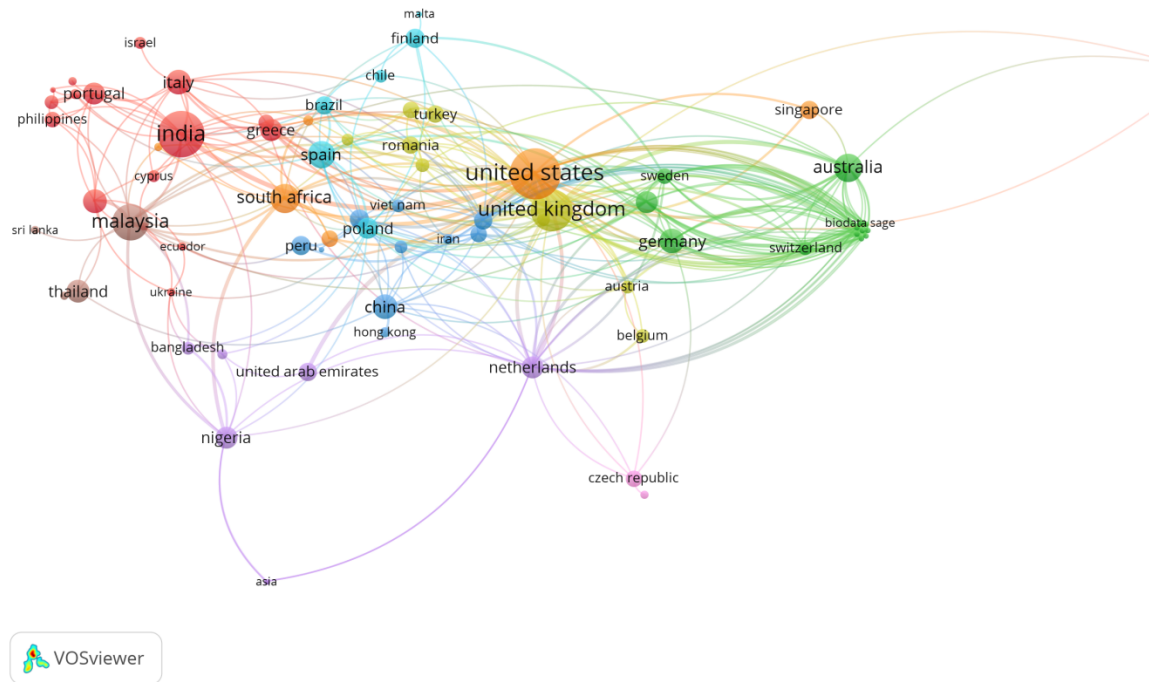


Figure 5: Country analysis.

This phenomenon also reflects the deep contradictions in the global knowledge production system. There is a structural mismatch between the “skill standard hegemony” constructed by developed countries through academic discourse power and developing countries’ needs for industrial transformation and upgrading.

Table 4: The top ten countries with the highest number of publications

Country	Documents	Citations
United States	79	711
India	65	430
United Kingdom	53	179
Malaysia	41	103
South Africa	25	152
Australia	24	249
Spain	22	140
Italy	19	121
China	18	74
Germany	18	102

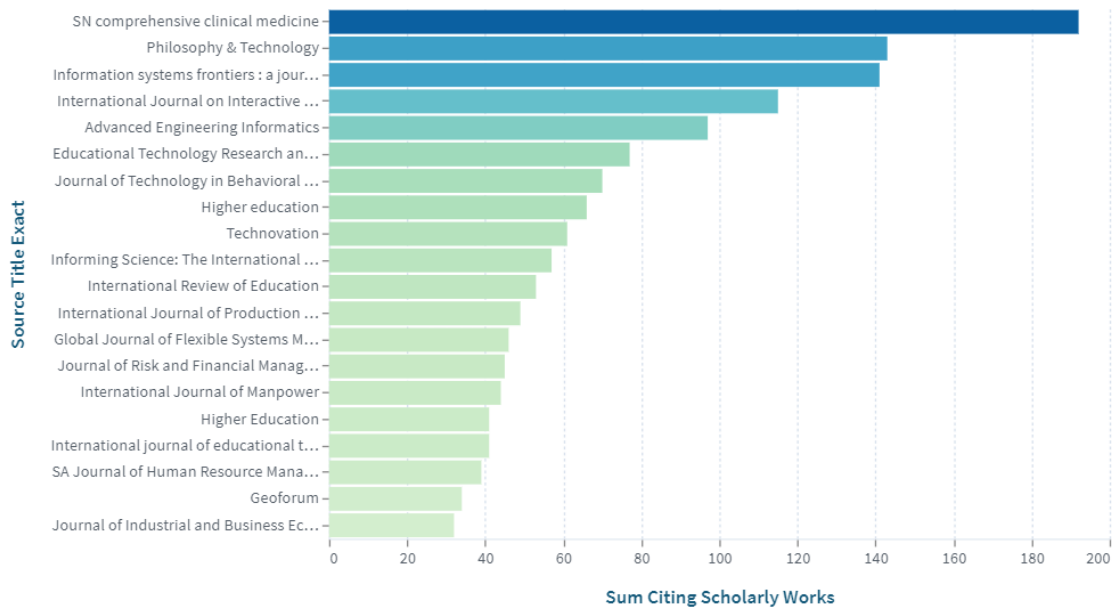


Figure 6: Top 20 frequently published journals from 2015 to 2025.

Figure 6 lists the top 20 journals with the highest citations and the subject area and category from 2015 to 2025. SN comprehensive clinical medicine (192), Philosophy & Technology (143), and Information Systems Frontiers: A Journal of Research and Innovation (141) are the three most-cited journals. These journals reflect the broad scholarly engagement with the topic, demonstrating that reskilling and upskilling are not only concerns in workforce development and education but are also increasingly relevant in healthcare, technological ethics, and digital innovation. The presence of journals from diverse subject areas suggests a growing cross-disciplinary convergence, where themes such as artificial intelligence, lifelong learning, and Industry 4.0 influence research agendas across multiple domains. This interdisciplinary visibility underlines the importance of skills development and its widespread applicability in the context of rapid technological transformation.

Figure 7 displays the top 20 productive institutions. The Universiti Teknologi MARA (UiTM) scored the highest, followed by the University of Technology Malaysia (UTM) in Malaysia, demonstrating their strong commitment to publishing and addressing the reskilling and upskilling research field. In the third position is Mahasarakham University (MSU) in Thailand, which shares the exact publication count as Pennsylvania State University, USA, placing them jointly in fourth place. Amity University, a prominent private institution in India, occupies fifth place. At the same time, the sixth place is held by Harvard University, underscoring the role of globally recognized institutions in contributing high-impact research to this domain. This ranking not only reflects institutional productivity but also suggests regional concentrations of expertise, particularly in Asia and North America, that are shaping the future of workforce development through technological adaptation, policy innovation, and reskilling/upskilling.

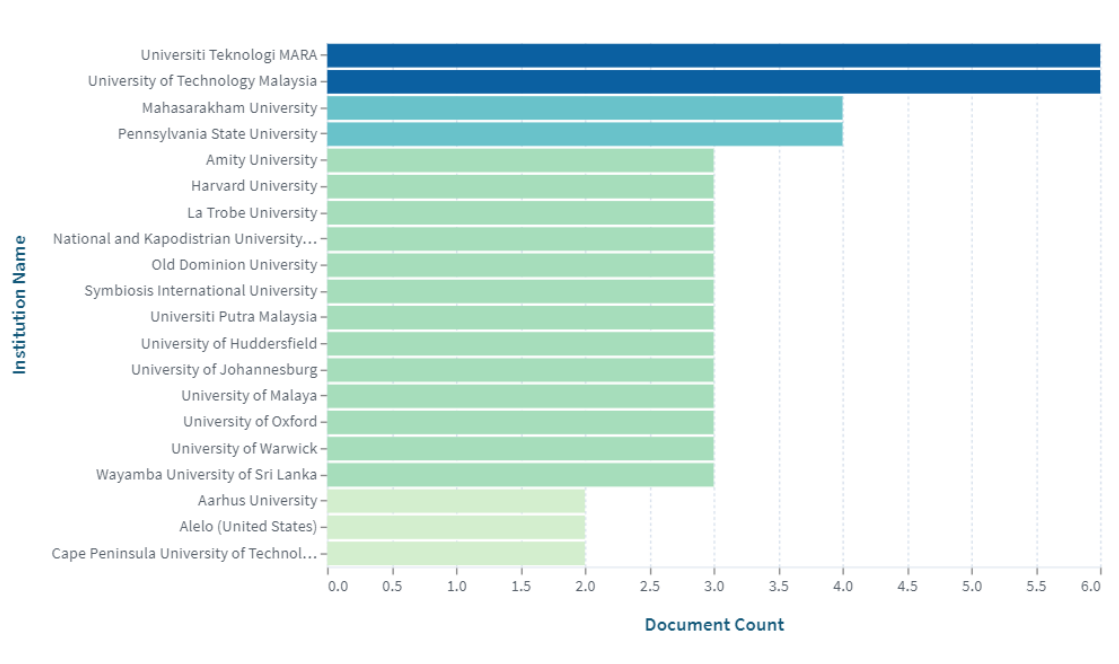


Figure 7: Top 20 productive affiliations from 2015 to 2025.

5. DISCUSSION

5.1 Paradigm Shift in AI Skill Development

From the perspective of the intersection of technological sociology and educational economics, these highly cited studies construct a two-way feedback loop of “skill shaping-technology adaptation”. They not only reflect the passive response of the academic community to the transformation of Industry 4.0 but also participate in the process of constructing future skill standards in the background. It is worth pondering that there are three key tensions in current research. First, the dispute over the right to define skills - when American scholars emphasize AI programming ability and German research focuses on the “craftsman intelligence” of human-machine collaboration, this competition in academic discourse is essentially a reconstruction of the global labor value system. Second, the educational time lag effect - the existing literature focuses on current technological needs, but rarely discusses the fundamental contradiction between the 5-10 years training cycle of the education system and the speed of technological iteration (such as the emergence of large language models). Third, the skill democratization paradox - although research calls for the improvement of digital skills for all, traditional industrial workers who need transformation the most are often excluded from academic dialogue. It is suggested that future research should: 1) develop a hysteresis model of “technology penetration-skill response” to quantify the half-life of skills in different industries; 2) establish a cross-national comparison framework to analyze the differences in the efficiency of vocational education systems in converting academic research results in various countries; 3) introduce the perspective of labor subjectivity and allow front-line workers to participate in the formulation of skill standards through action research. Only by breaking through

the “technological determinism” paradigm of existing research can we truly achieve a paradigm shift from “human capital adapting to technology” to “technological development serving human value”.

5.2 AI Skill Enhancement Practices in Malaysia

The core technologies of Industry 4.0 - automation, artificial intelligence (AI) and big data [45] - are reshaping the focus areas of human resource management. While traditional research has focused on static capacity building such as employee training [46], knowledge management [47], and learning and development (L&D) [48], current research is focusing on the more dynamic concepts of skills retraining (reskilling) and skills enhancement (upskilling). This research shift stems largely from the real-world challenges of accelerating technology iterations [49]. In the AI-driven workplace environment, employees must continuously update their skills to remain competitive [50, 51]. Therefore, scholars generally emphasize that continuous learning and development is not only the key to individual career growth, but also a core mechanism for organizations to adapt to technological change [52–54].

Malaysia is actively promoting AI skills upgrading and retraining to address the need for workforce transformation due to technological change. The government is promoting AI literacy for all through large-scale programs such as the “AI for Malaysia’s Future” program with Microsoft (targeting to train 800,000 Malaysians) and the Google-backed “AI at Work 2.0” program (training 445,000 civil servants) to enhance AI literacy for all [55, 56]. In addition, the MyMahir National AI Industry Council (MyMahir-NAICI) was established to provide retraining and skills upgrading support to AI-affected industries, with an estimated 620,000 jobs expected to be impacted by AI in the next 3 to 5 years [57]. Meanwhile, the National Training Week (NTW) was extended to ASEAN, offering 65,000 free courses covering areas such as AI and digitalization. These initiatives show that Malaysia is building an AI-ready labor market through public-private partnerships, education and policy coordination [58]. For example, Universiti Putra Malaysia (UPM) has adopted a multi-dimensional strategy for AI skills development, focusing on a combination of academic research, industry collaboration and community service, such as automation and robotics management [59]. Universiti Teknologi Malaysia (UTM) has established the Center for Artificial Intelligence (CAIRO) to promote interdisciplinary research in areas of Malaysia’s strengths, such as agriculture and health-care, etc [60]. Universiti Sains Malaysia (USM) has been selected as one of the universities in Malaysia to participate in the Artificial Intelligence in Universities (AI4U) program, which is jointly organized by the Malaysian Investment and Development Authority (MIDA), Intel Corporation and Malaysian Production Corporation (MPC) [61]. In 2025, Universiti Kebangsaan Malaysia (UKM) also develops AI learning modules through the Artificial Intelligence Summer Camp (AISC) program, such as chatbot development [62].

6. CONCLUSION

Globally, AI skills upgrading and retraining have become a core strategy for dealing with technological change. Organizations generally prefer to adapt to AI transformation through training rather than layoffs. The bibliometric analysis of this paper based on VOSviewer software reveals significant academic value in the field of reskilling and upskilling research. Articles analyzed vary

widely in their coverage of year of publication, type of article, source. It provides a review of the primary literature in the field and offers insights into the current state of development. The study found that the research on AI skills upgrading and retraining shows a rapid growth trend, with journal articles focusing on the following directions. Firstly, from the perspective of time dimension, the relevant studies show obvious stage characteristics. 2015-2018 is dominated by the construction of theoretical frameworks, while after 2019, it shifts to the exploration of application scenarios, especially the research on new AI-driven training modes grows significantly. Second, the distribution of literature types shows a higher proportion of empirical studies, but there are insufficient cases involving developing countries, exposing the geographical imbalance of research. Third, the source of literature shows that the highly cited literature is mostly concentrated in 10 core journals such as SN comprehensive clinical medicine, forming an obvious “core-marginal” journal distribution structure.

With respect to the content of the study, the first is to explore the structural impact of AI on the labor market, including the risk of job substitution and the demand for emerging skills. The second is to study the application of AI in personalized learning, such as the use of AI in the field of higher vocational education to dynamically adjust the learning path and improve the efficiency of students’ skill mastery. Third, it analyzes the collaboration model of government, enterprise, and academia, such as the key role of industry alliance and non-profit organization cooperation in retraining. In terms of research methodology, recent journal articles combine quantitative and qualitative analyses, including bibliometrics (e.g., VOSviewer analysis), case studies (e.g., evaluation of corporate training programs), and experimental studies (e.g., comparison of AI personalized learning effects). Current research gaps include long-term training effectiveness tracking, accessibility assessment for vulnerable populations, and the impact of AI-assisted training on traditional skills retention.

While this research provides a complete overview for practitioners and scholars, it has some limitations. The study was conducted on articles indexed in the Scopus database and didn’t include other pertinent databases, such as ACI (ASEAN Citation Index), MyCite (Malaysian Citation Index), SciELO (Scientific Electronic Library Online). Further research could integrate qualitative methods like content analysis or expert interviews, which would supplement the bibliometric insights with context-rich analyses of how reskilling and upskilling are undertaken in various sectors. It is particularly noteworthy that this study systematically reveals for the first time the “center-periphery” structure of global skills research. There is a clear gap between the theoretical framework dominated by European and American scholars and the local practices in developing countries. This geographical inequality in knowledge production has forced Africa, Latin America and other regions to apply Western countries’ skill solutions often when dealing with the labor market impact brought by AI, while ignoring the particularities of the local industrial ecology.

Conflict of interest

There are no conflicts of interest in this study.

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